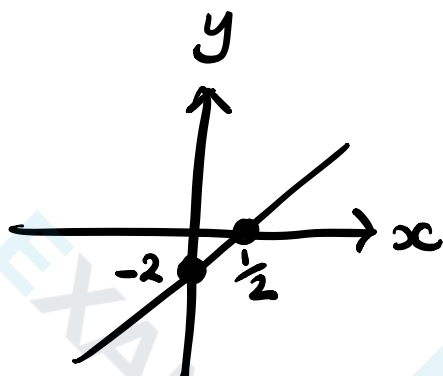


Graphs of Modulus and Reciprocals

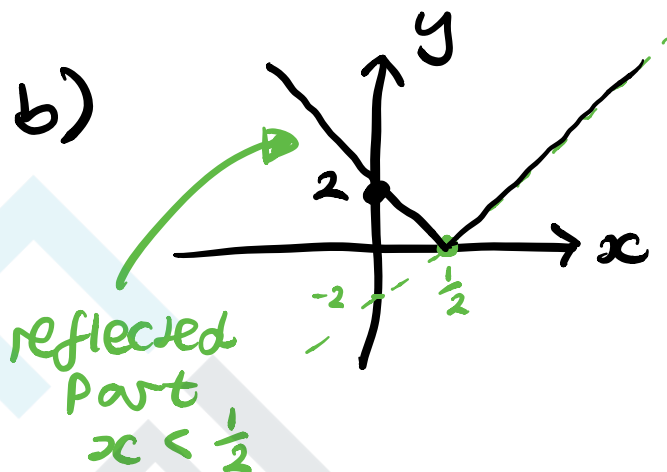
Basic skills

Q1.

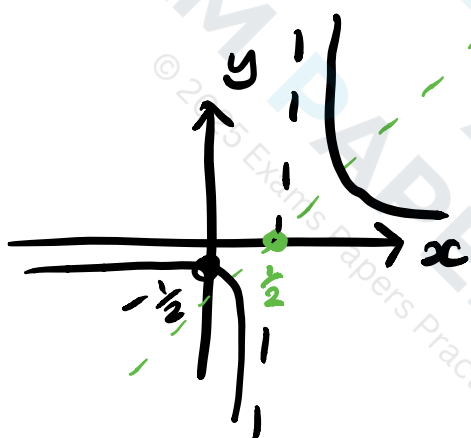
a)



b)

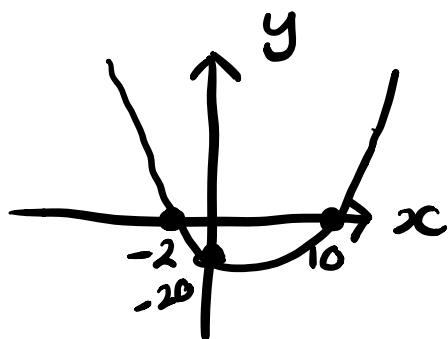


c)



Q2.

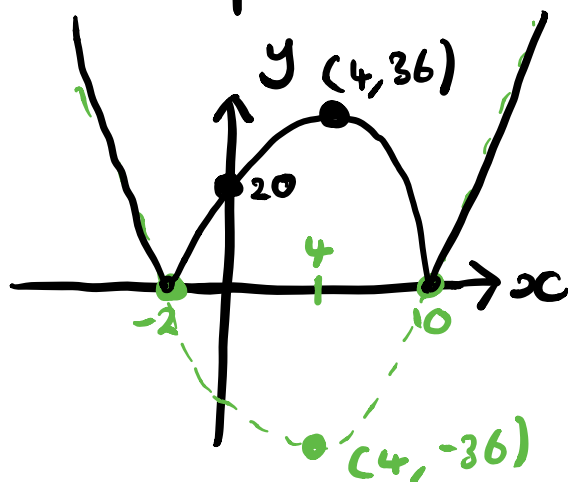
a)



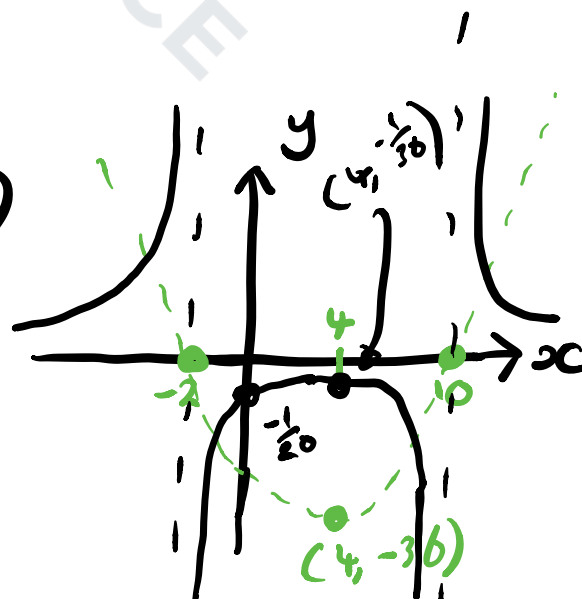
$$(x-10)(x+2)$$

roots
 $x = -2, 10$

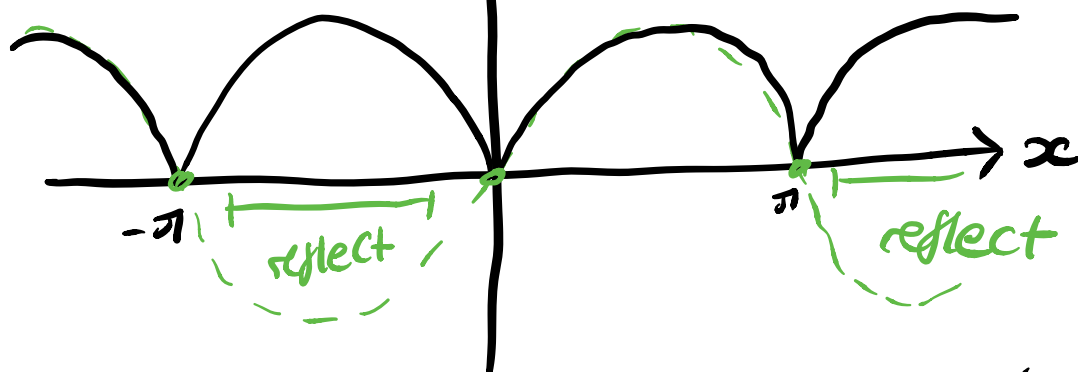
b)



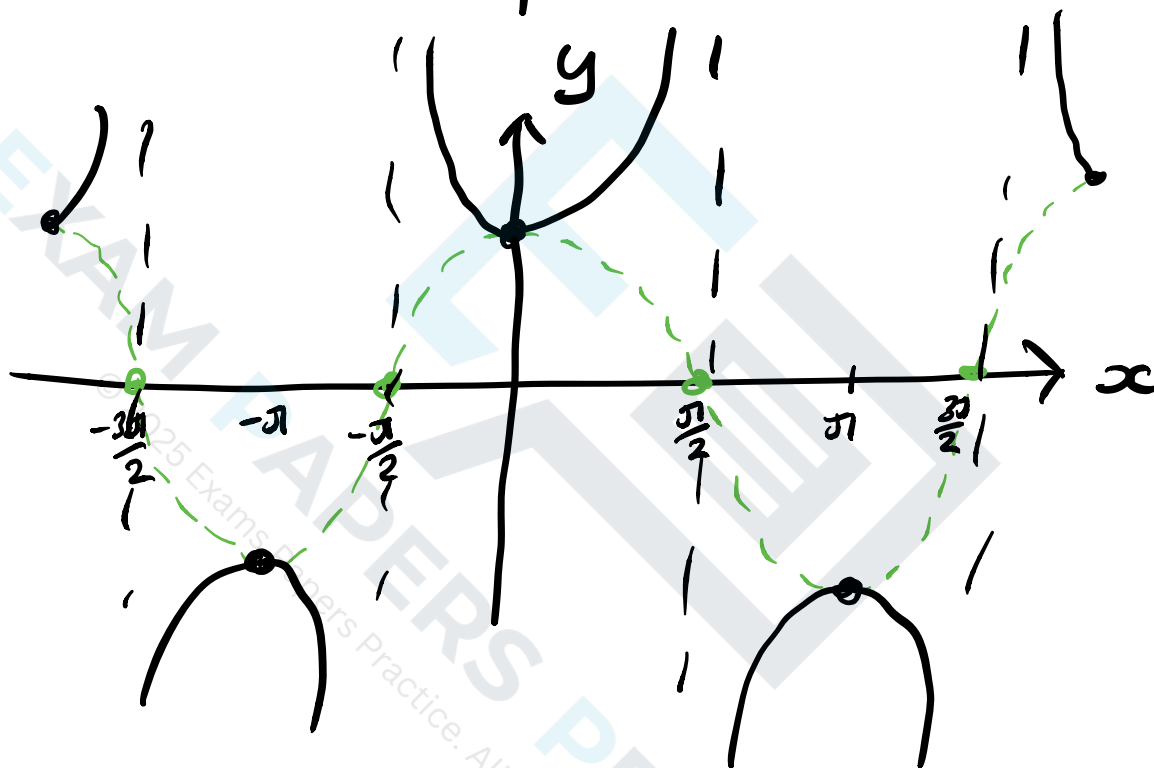
c)



Q3.



Q4.



Competency skills

Q5. a) For intersection solve $|2-3x|=x+5$

so

$$2-3x=x+5$$

with $2-3x > 0$ i.e. $x < \frac{2}{3}$

$$4x = -3$$

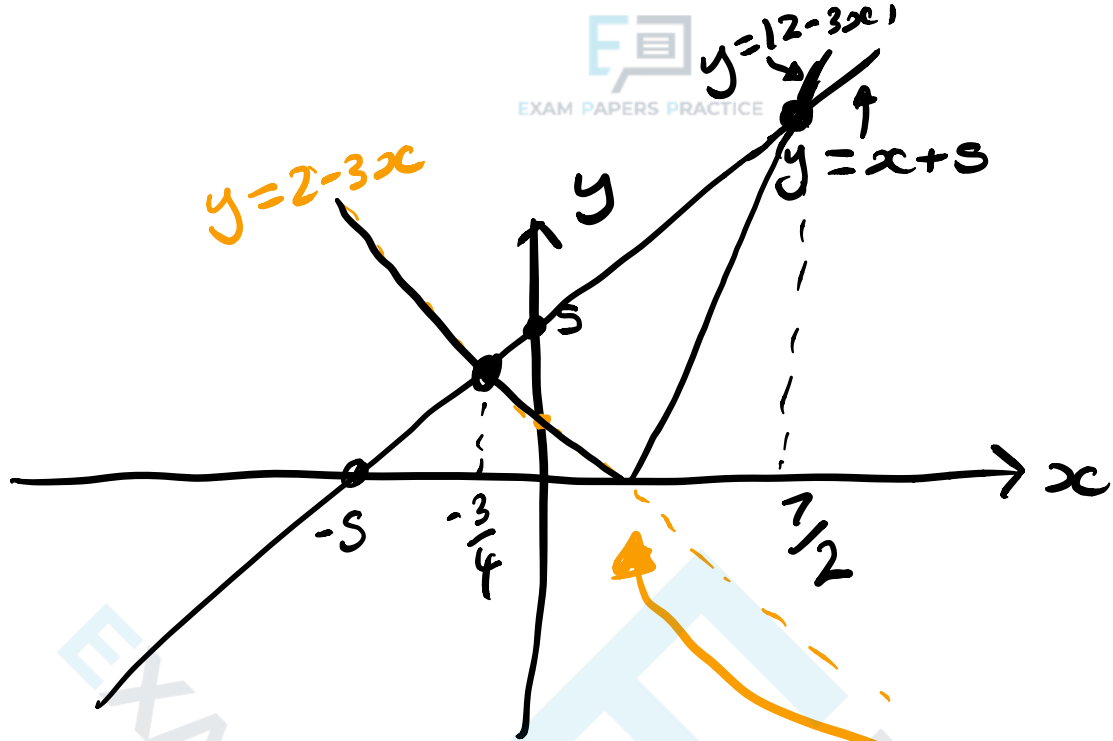
$$x = -\frac{3}{4}$$

$$3x-2=x+5$$

with $x > \frac{2}{3}$

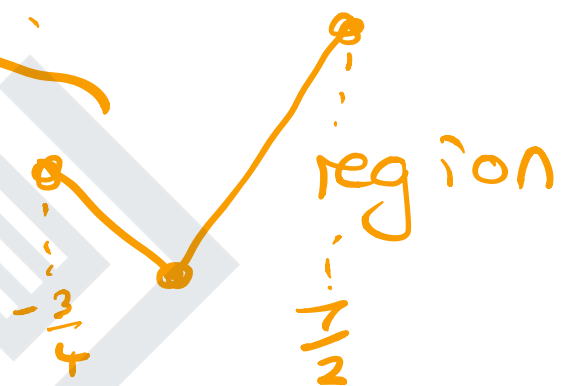
$$2x = 7$$

$$x = \frac{7}{2}$$

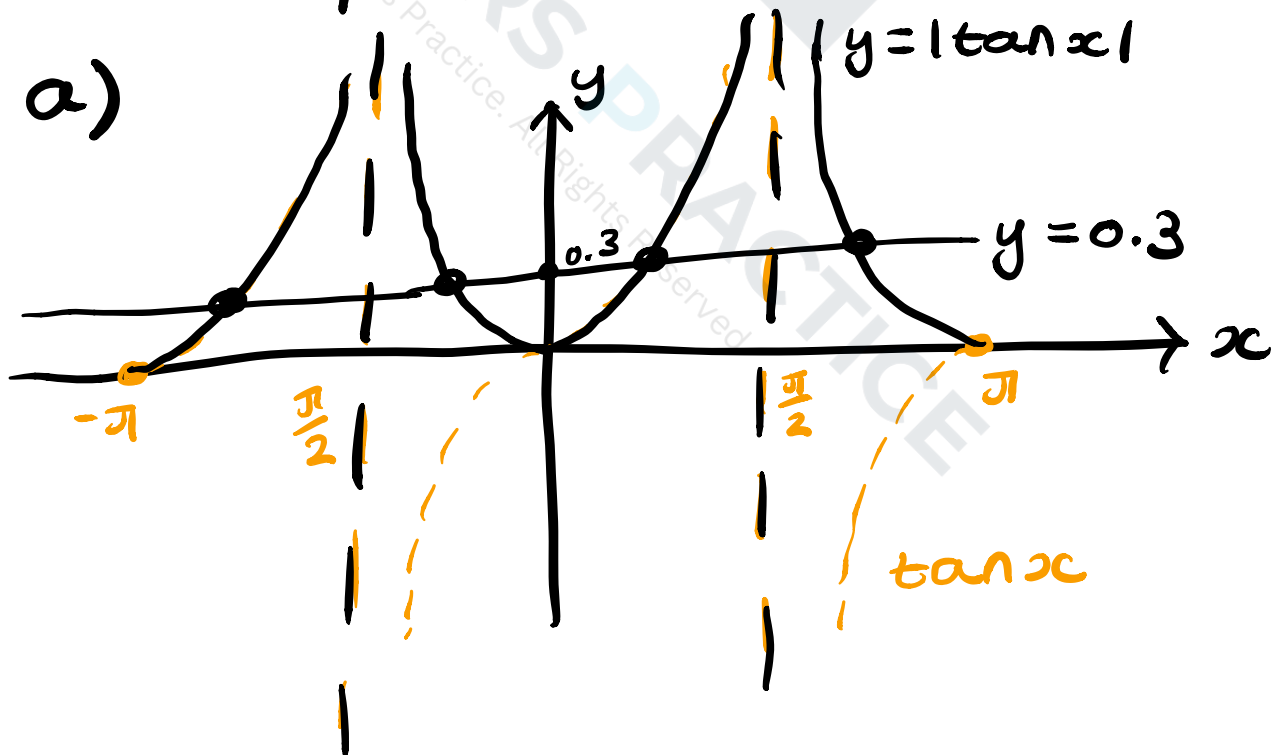


b) $|2 - 3x| < x + 5$

so $-\frac{3}{4} < x < \frac{7}{2}$



Q6. a)



b) $|\tan x| = 0.3$ to find intersection points on graph

$$\tan x = 0.3$$

$$0 < x < \frac{\pi}{2},$$

$$-\pi < x < -\frac{\pi}{2}$$

$$-\tan x = 0.3$$

$$-\frac{\pi}{2} < x < 0$$

$$\frac{\pi}{2} < x \leq \pi$$

$$x = \tan^{-1} 0.3$$

$$= 0.29$$

$$\text{and } \downarrow -\pi \text{ for sol}$$

$$x = 0.29 - \pi$$

$$x = \tan^{-1}(-0.3)$$

$$= -0.29$$

$$\downarrow +\pi \text{ for sol}$$

$$x = \pi - 0.29$$

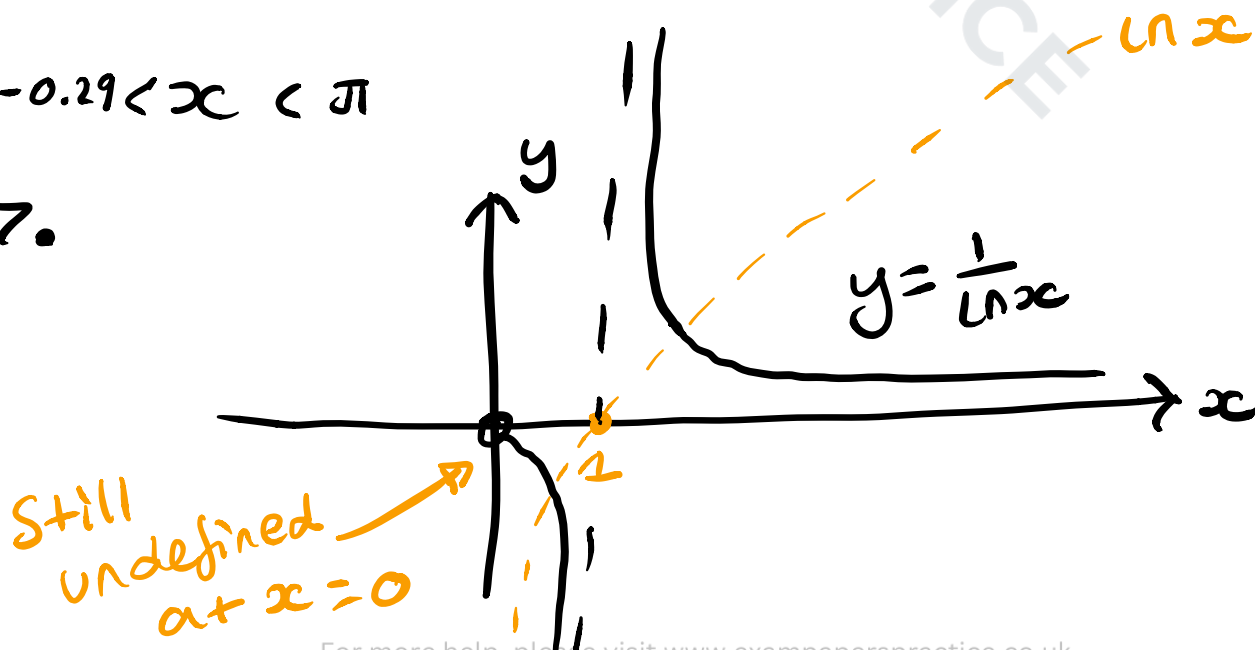
so for inequality



$$-\pi < x < 0.29 - \pi, \quad -0.29 < x < 0.29,$$

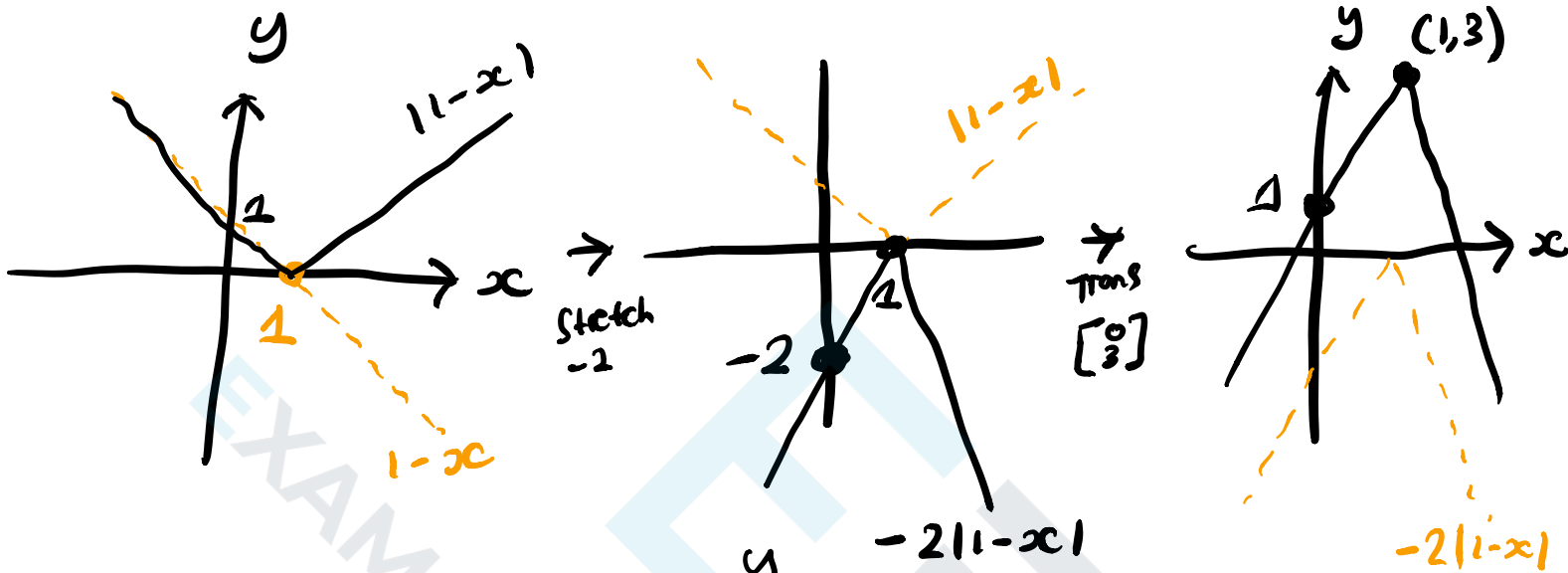
$$\pi - 0.29 < x < \pi$$

Q7.



Q8. For $y = 3 - 2|1-x|$ first plot

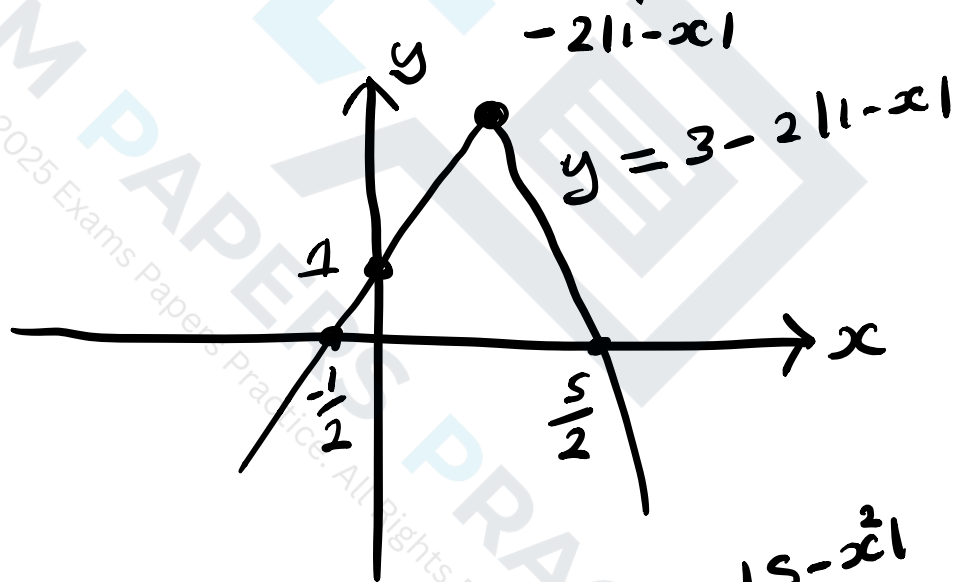
$$|1-x| \rightarrow -2|1-x| \rightarrow 3-2|1-x|$$



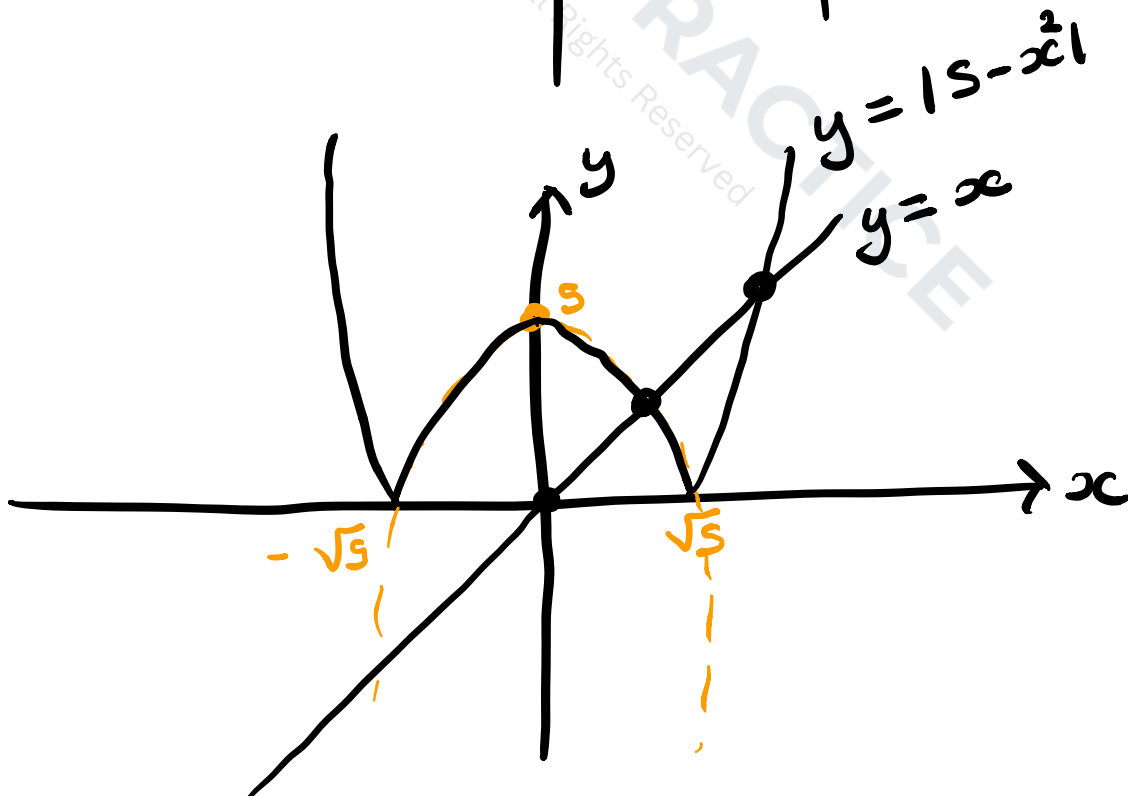
For x-int:

$$|1-x| = \frac{3}{2}$$

$$x = -\frac{1}{2}, \frac{5}{2}$$



Q9.



For intersection solve $|5-x^2| = x$

so $5 - x^2 = x$

for $-\sqrt{5} < x < \sqrt{5}$

solve

$$x^2 + x - 5 = 0$$

$$x = \frac{-1 \pm \sqrt{21}}{2}$$

so $x = \frac{-1 + \sqrt{21}}{2}$

✓ in domain

$$x^2 - 5 = x$$

$$x < -\sqrt{5}, x > \sqrt{5}$$

solve

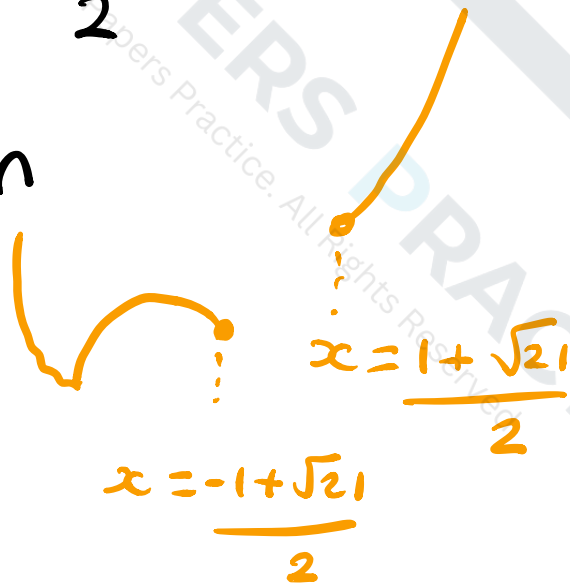
$$x^2 - x - 5 = 0$$

$$x = \frac{1 \pm \sqrt{21}}{2}$$

✓ in domain

$$x = \frac{1 + \sqrt{21}}{2}$$

so region

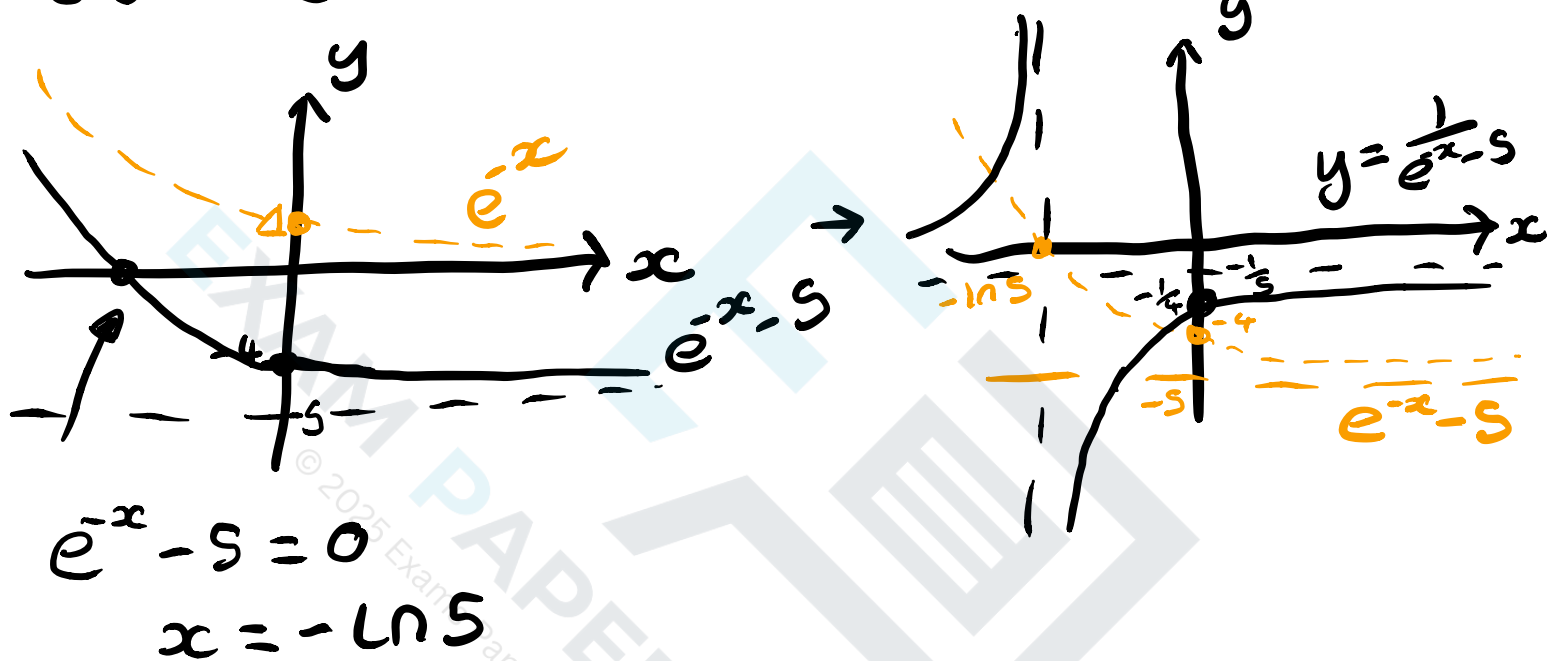


so

$$x < \frac{-1 + \sqrt{21}}{2} \quad \text{or} \quad x > \frac{1 + \sqrt{21}}{2}$$

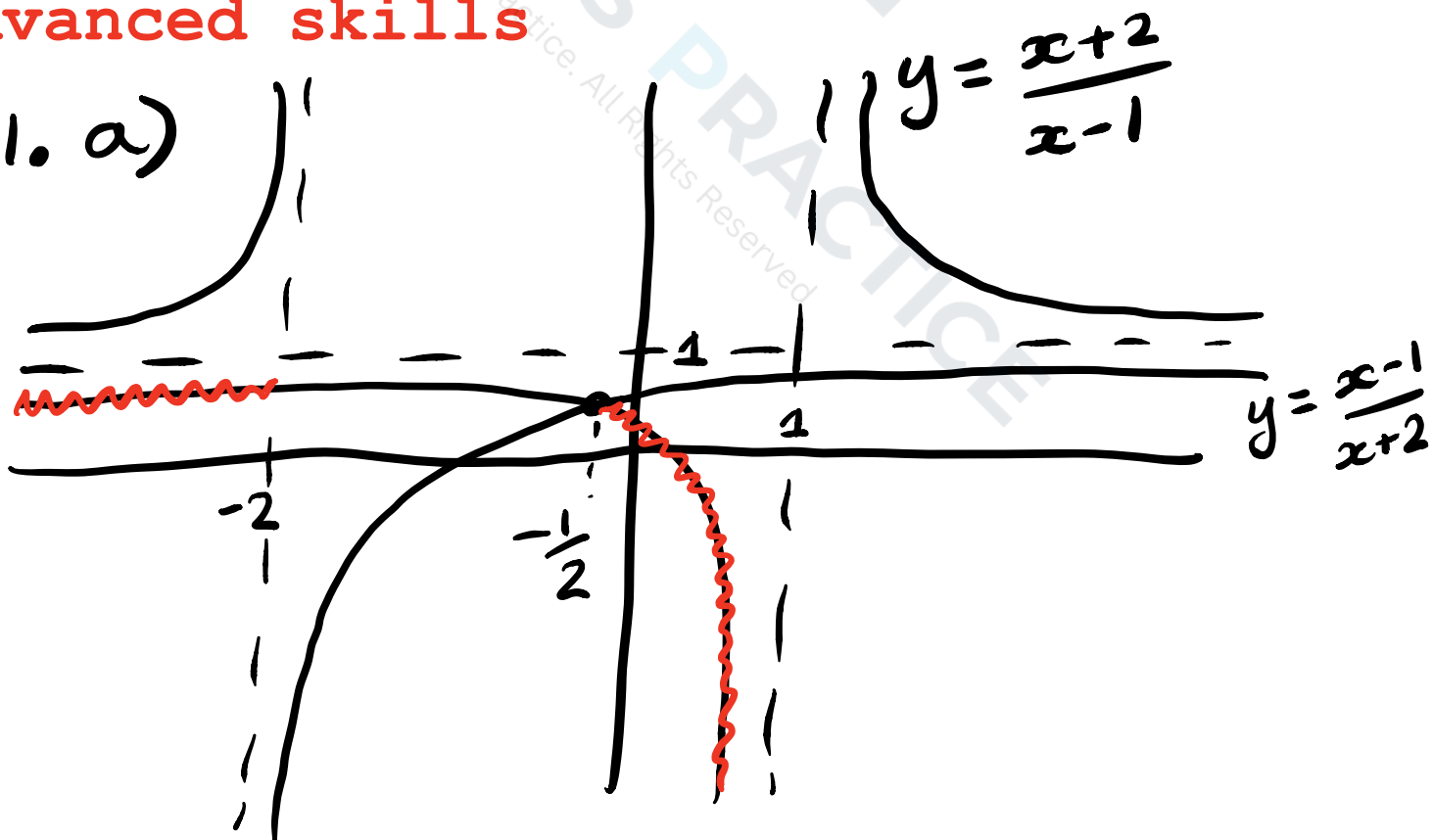
Q10. $y = \frac{1}{e^{-x} - 5}$

Plot $e^{-x} - 5$ then reciprocal



Advanced skills

Q11. a)



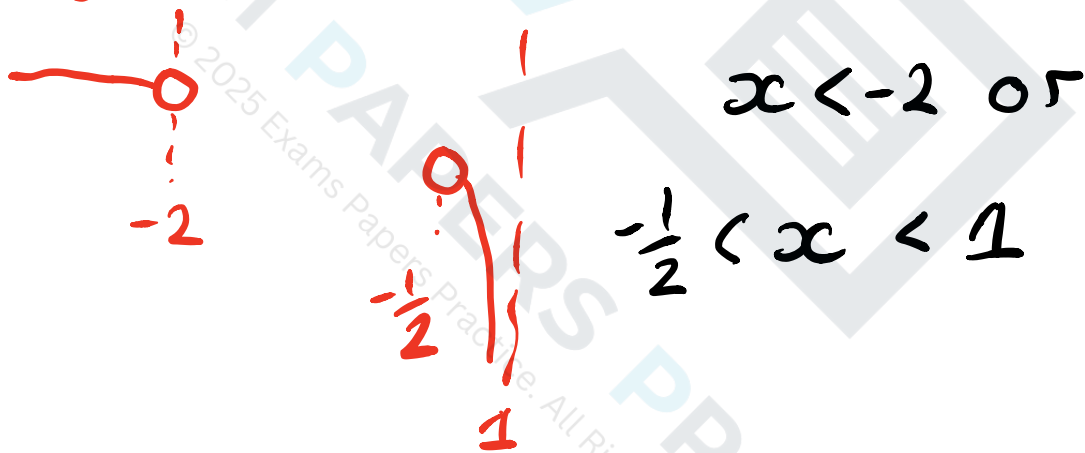
For intersection

$$\frac{x+2}{x-1} = \frac{x-1}{x+2} \quad (x+2)^2 = (x-1)^2$$

$$x^2 + 4x + 4 = x^2 - 2x + 1$$

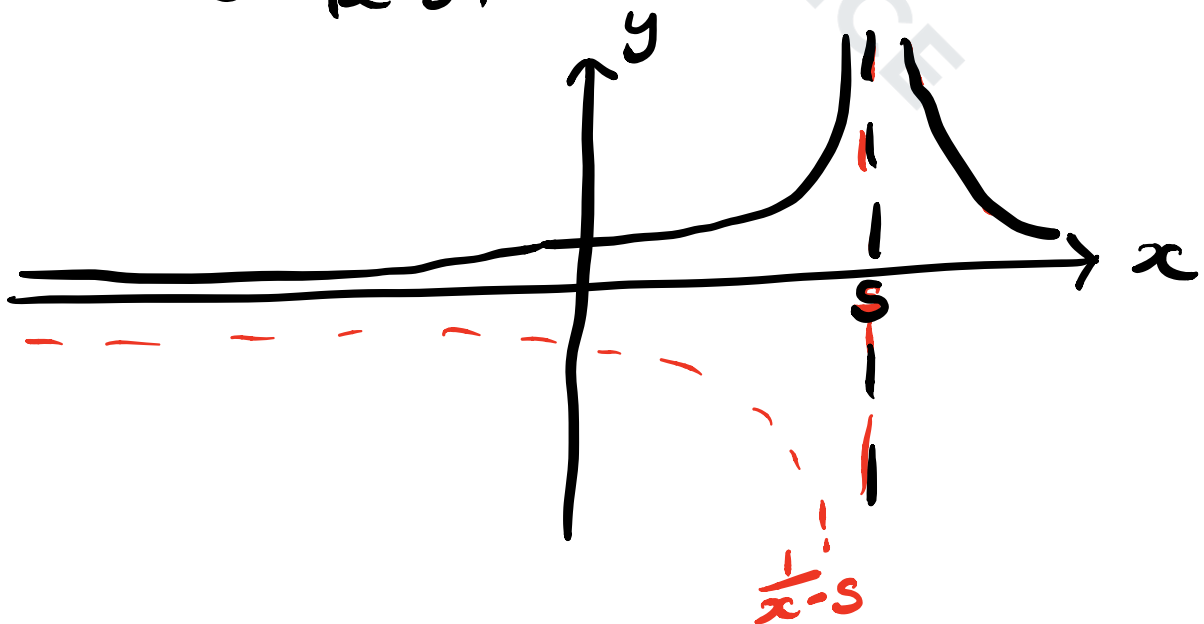
$$6x = -3 \quad x = -\frac{1}{2}$$

b) Highlight on diagram region



Q12.

TO Plot $y = \left| \frac{1}{x-5} \right|$ Plot $\frac{1}{x-5}$ take modulus



Q13. $y = a|e^{2x} - 3| + b$

fit $(5, -5)$: $-5 = a|e^{10} - 3| + b$
 $-5 = a(e^{10} - 3) + b$

fit $(0, 2)$: $2 = a|e^0 - 3| + b$
 $2 = a|-2| + b$
 $2 = 2a + b$

so $b = 2 - 2a$

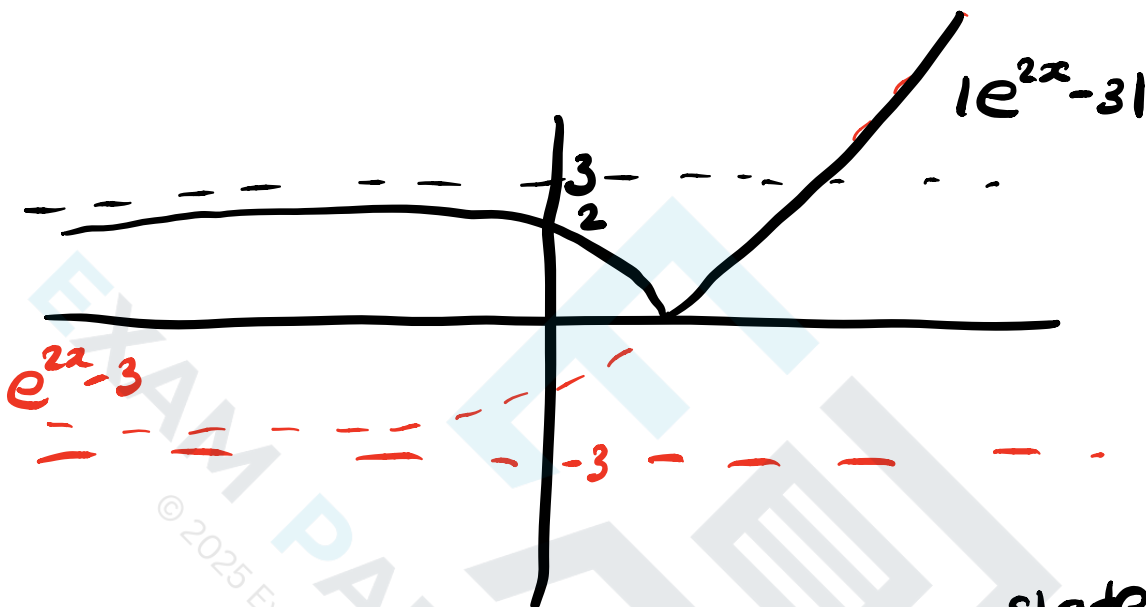
so $-5 = a(e^{10} - 3) + 2 - 2a$

$-7 = a(e^{10} - 5)$

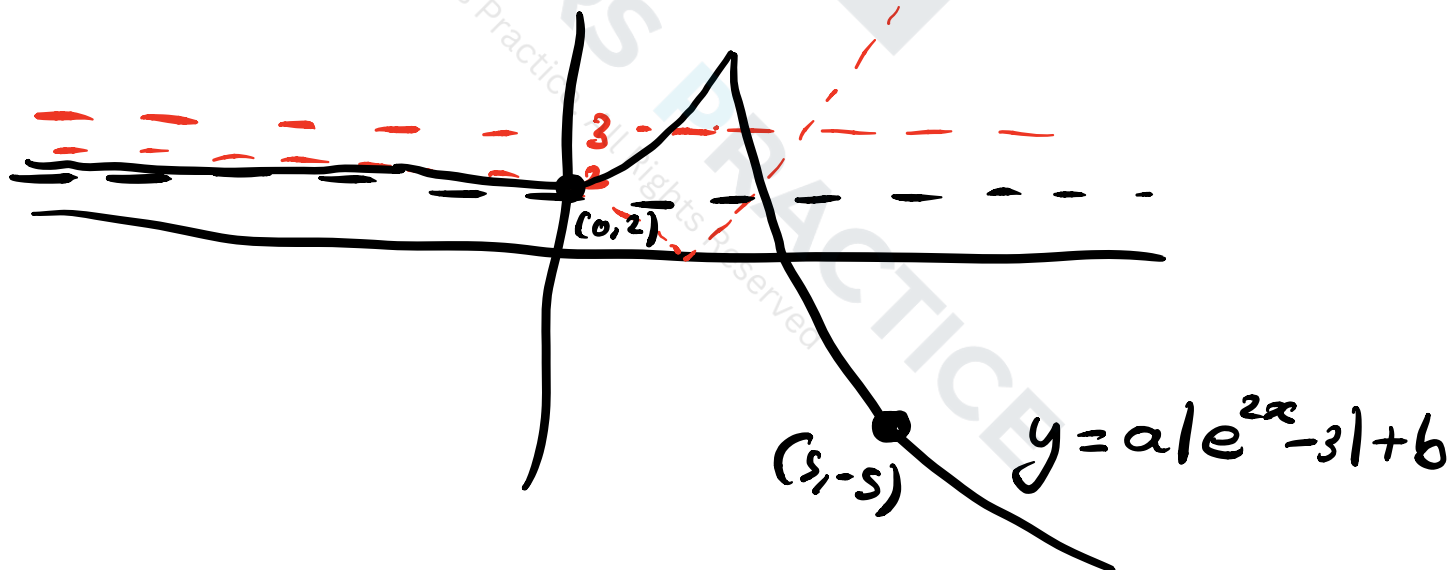
$a = \frac{-7}{e^{10} - 5}$ $b = 2 + \frac{14}{e^{10} - 5}$

$a = \frac{7}{5 - e^{10}}$ $b = \frac{2e^{10} + 4}{e^{10} - 5}$

so $y = \frac{7|e^{2x}-3|}{5-e^{10}} + \frac{2e^{10}+4}{e^{10}-5}$

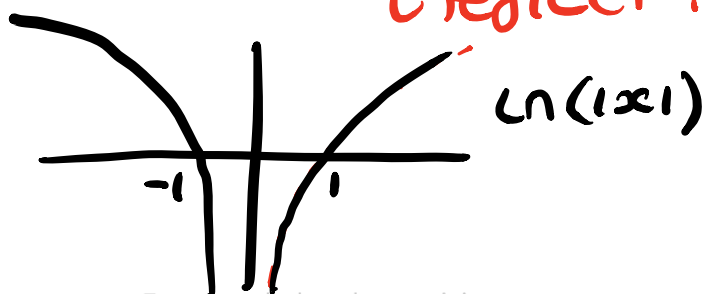


↓ scale and translate
Note: $a < 0$

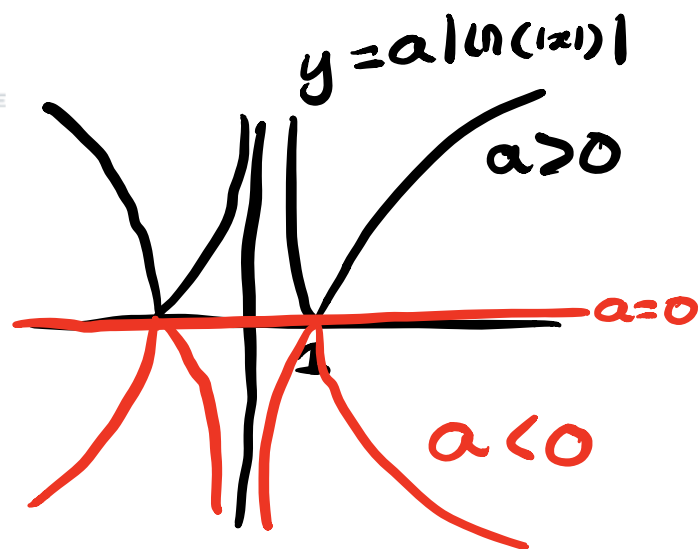
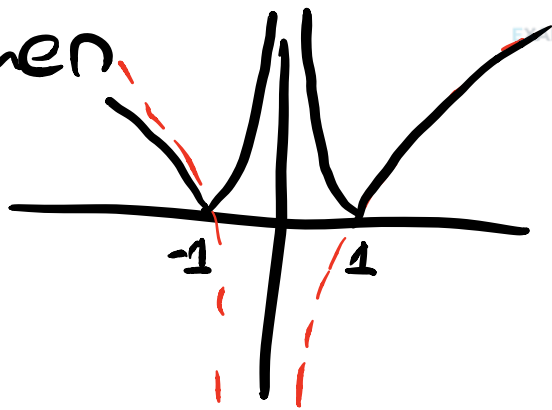


Q14. $y = a|\ln(|x|)|$

↑ reflect in y-axis (input $\pm$ ignore)

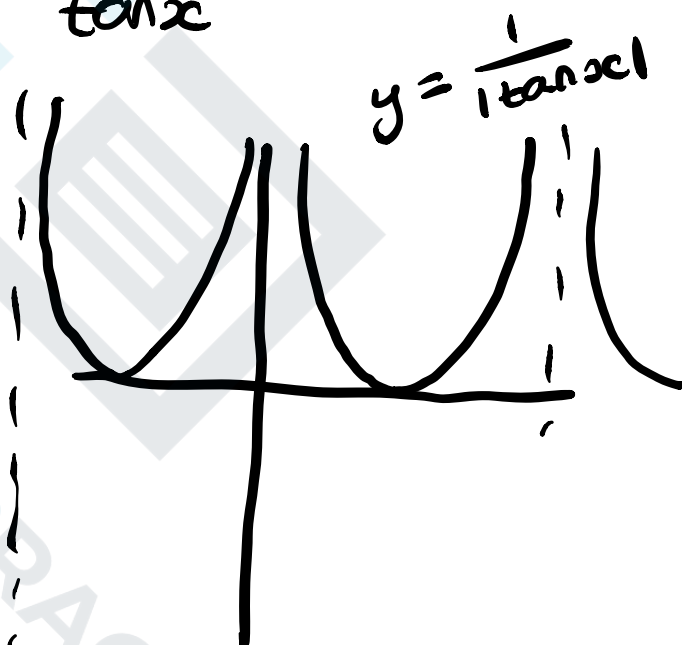


then



Q15. $y = \frac{1}{|\tan x|}$

plot $\frac{1}{\tan x} = \cot x$



yes $\frac{1}{|\tan x|} = |\cot x|$

As

$$\left| \frac{1}{\tan x} \right| = \frac{1}{|\tan x|}$$